

PRE-CHEM KINETICS

Energetics of a reaction:

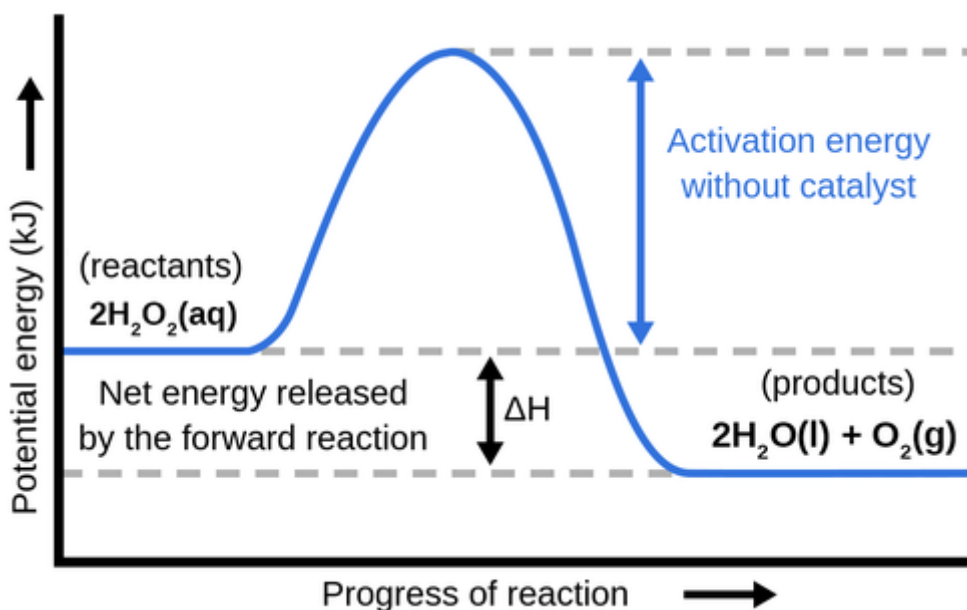
1. Exothermic - Energy released (drop in the PE diagram). Bonds formed are exothermic.

$$\Delta H < 0$$

2. Endothermic - Energy absorbed. Bonds broken are endothermic. $\Delta H > 0$

Activation energy: Energy to spark the reaction, from the reactants to the transition state (for an overall reaction's energy curve, it's reactant to intermediate). For a mechanism, you'll see multiple humps here, each step has its own activation energy, and the rate determining step (slow step) will likely be the one with the highest activation energy.

What is the below diagram?



Whether the reaction can get over the hump, a big determiner in the activation energy depends on:

1. Frequency of collisions (Can manipulate this via T, which is average KE - if T increases, k increases)

1. Arrhenius equation: $k = Ae^{\frac{E_a}{RT}}$, where A is a pre-exponential factor

2. Energy and orientation of the molecules (sterics) - these determine A
3. Presence of a catalyst - it decreases E_a .

Rates: Change in concentration over time (units: $M/s = mol L^{-1} s^{-1}$).

1. Average rate = $\frac{\Delta C}{\Delta t}$. (Slope of secant line of C vs t)
2. Instantaneous rate = $\frac{dC}{dt}$. (Slope of tangent line of C vs t)

Rates depend on stoichiometry - take an example $2A + B \rightarrow 3C$. Say the rate of decrease in A is some value x. B decreases by 0.5x, and then C is generated at a rate of 1.5x.

Equilibrium: Forward and reverse reaction rates are equal, net change in concentration is 0 (C vs t plot stabilizes). Remember that $K = \frac{k_f}{k_r}$, and the tendency of equilibrium depends on the rates of forward and reverse reactions. So, by extension, if $k_f > k_r$, $K > 1$, and $\Delta G < 0$, leading to spontaneity.

LeChatlier's principle: With a perturbation, a reaction will tend towards equilibrium:

1. Increase the concentration (or pressure) of a reactant, reaction tends to the right towards products, increase pressure of product, reaction tends left towards reactants.
2. Change volume: Pressure will change, reaction will tend towards lower pressure if volume is decreased, higher pressure if volume is increased.
3. Change T: This affects the equilibrium constant, for an exothermic reaction, shift left (by decreasing K), for an endothermic reaction, shift right (by increasing K). You can actually use T to kick start a reaction into spontaneity, you increase T, increasing K which decreases G. Both defn of G support this: $\Delta G = \Delta H - T\Delta S = -RT\ln K$.

Reaction mechanisms: Multiple steps to an overall reaction. Has reactants and products like an overall reaction, but there is also the presence of the reactive intermediate, which is produced in one step and consumed in another. Catalysts or collision partners are present in reactant and product.

Reaction order demonstrates how rate varies with concentration of a reactant. Let's say there's some reaction $A + B \rightarrow C$, $r = k[A]^n[B]^m$, where n is the order of A, and m is the order of B. Overall order is n+m.

1. Zero order - No dependence of rate on concentration. $C - C_0 = -kt$, so concentration is linear with t, with a slope of -k. k's units are $M s^{-1}$
2. First order - Rate scales linearly with concentration. $\ln(C) - \ln(C_0) = \ln(\frac{C}{C_0}) = -kt$, concentration decays exponentially with t, slope of $\ln(C)$ with t is -k. k's units are s^{-1}
3. Second order - Rate scales quadratically with concentration. $\frac{1}{C} - \frac{1}{C_0} = kt$. $\frac{1}{C}$ vs t is linear, with slope of k. Units of k are: $M^{-1} s^{-1}$.

Note that the units of k are dependent on the OVERALL reaction order.

Method of initial rates:

Purpose:

1. Elucidate reaction orders experimentally
2. Validate guesses of reaction mechanisms
3. Calculate reaction rate constant

Process:

1. Measure initial rates while varying concentrations of one reactant at a time. (Ideally double, triple, some easy factor). Note - all experiments should be done at the SAME ambient conditions.
2. Check the multiplicative effect on rate constant to get order (e.g., if r quadruples when [A] doubles, 2nd order in A)
3. Once the rate expression is complete, choose one representative run and plug in r, C's to get k. Order determines k's units.

MASS BALANCES

$$In - Out + Gen - Cons = Accumulation$$

$$\frac{dN_j}{dt} = F_{j0} - F_j + G_j$$

Appropriate simplifications: $\frac{dN_j}{dt} = \frac{dC_j V}{dt}$, and if V is constant, it equals: $V \frac{dC_j}{dt}$

$G_j = \int r_j dV$, and assuming perfect mixing: $G_j = r_j V$.

S.S.: $\frac{dN_j}{dt} = 0$, Batch: $F_{jo} = F_j = 0$.

Solving rate problems:

1. Write out the material balance
2. Write out the rate equations, and initial conditions. Verify each assumption!!
3. Look for equalities, relations, etc... (how can we express C_A in terms of C_B , etc...)
4. Based on the DE's form, see what solving method is needed! (Separable, integrating factor, matrix+eigenstuff (when unable to decouple but can express as a system))
5. THINK ABOUT RELATIVE VALUES OF k TO SIMPLIFY. The entire (perceived) order of the rxn can change based on these relations. Also, $K = \frac{k_f}{k_r}$ could be useful for certain situations
6. Solve

STOICHIOMETRY

Keep in mind: for a reaction $aA + bB \rightarrow qQ + sS$

$$-\frac{1}{a} \frac{dN_A}{dt} = -\frac{1}{b} \frac{dN_B}{dt} = \frac{1}{s} \frac{dN_S}{dt} = \frac{1}{q} \frac{dN_Q}{dt}$$

Can write out multiple reactions in matrix form:

Stoichiometric matrix has reactions in the rows, species in the columns.

$$\frac{dC}{dt} = Kc, \text{ where } K \text{ is a matrix of rate constants, rxns on rows, species on columns}$$

Linear independence of reactions can be gauged using linear algebra (for a square matrix, determinant can be used, and rank can be used no matter what)

1. Reduce to REF (Row-echelon form)
2. Rank = # of linearly independent reactions

Boolean matrix (incident matrix): Rows are species, columns are reactions (reverse and forward are separate), values are produced (1), consumed (-1), or not involved (0). Boolean sum is the net. It is nonzero only if the species is only produced or consumed.

Reaction mechanisms:

Concept of the reactive intermediate explains reactions where the stoichiometric reaction doesn't tell the whole story.

Elementary reaction: Chemical reaction that doesn't pass through stable intermediates. Can use mass action (coefficients and reactants determine the rate law as written), so for some step, $A + B \rightarrow C, r = k[A][B]$.

Ones that has only one reactant/product are impossible. (Unimolecular steps like $A \rightarrow B$ require a collision partner: $A + X \rightarrow B + X$), tertiary reactions are very improbable.

RDS: Rate-determining step - the overall rate is dominated by the rate of a single slow step.

PSSA approximation: When an intermediate is active, it reacts immediately ($k_2 \gg k_1$), and is very short lived. The net rate of that intermediate can be assumed to be 0, due to a pseudo-equilibrium.

LCA approximation: For radical mechanisms, propagation steps are much faster than initiation and termination.

QE approximation: Forward and reverse reactions are really fast, concentrations are related by thermodynamics. Validity is only there when these rates are much faster than the rate of the next reaction. $r_{-1} \gg r_2$.

MAI: Most abundant surface intermediate - the rate dependence on a reactant can change due to something that isn't a change in the mechanism, specifically a change in the most abundant surface intermediate.

De Donder:

$$\frac{\vec{r}}{\overleftarrow{r}} = \frac{\prod_i r_i}{\prod_i r_{-i}}$$

Essentially, for an overall reaction to be reversible, all r_{-i} should be nonzero, and to have a forward rate, all r_i should be nonzero.

Relate this with energetics: De Donder said that the chemical affinity: $A_i = -\Delta G_i$. This is 0 only at equilibrium, otherwise positive. By thinking of this with equilibrium, you get that $\frac{r_i}{r_{-i}} = e^{\frac{A_i}{RT}}$. Think abt this, what happens at equilibrium?

A: $A_i = 0$, and we get to a very familiar property, $r_i = r_{-i}$ at equilibrium.

De Donder's final result:

$$\frac{\vec{r}}{\overleftarrow{r}} = \frac{\prod_i r_i}{\prod_i r_{-i}} = e^{\frac{A_{rxn}}{\bar{\sigma}RT}}$$

Where $\bar{\sigma}$ is the affinity-weighted stoichiometric coefficient. Why important? Allows for evaluation of mechanisms.

When dealing with mechanisms and elucidating likely steps, consider:

1. E_a dependence
 1. Bond energies
 2. Radicals are unstable and have a tendency to react.
2. Preexponential factor dependence (collision theory)
 1. How do the sterics play a role?
3. Evans-Polanyi rule - for initiation step: $E_a = \Delta H_{rxn} + 1 \text{ kcal/mol}$

Stoichiometric matrix vs incident matrix (incident matrix is binary, consumed -1, produced +1, neither 0). Allows understanding on a simple boolean level.

Rate Constant Estimation

Two prevailing ways to figure out a rate constant from scratch:

1. Collision theory - Hard sphere model, and it represents the fact that collisions need sufficient orientation, energy, and frequency to interact. Temperature increases average KE, increasing movement. Larger collision diameter increases likelihood of favorable collisions, and steric factor also matters here.

From collision theory, we know that:

$$k = L \frac{\rho (\pi d_{AB}^2)}{4} \sqrt{\left(\frac{8RT}{L\pi\mu} \right)} e^{\frac{-E_a}{RT}} e^{\frac{1}{2}}$$

ρ is the steric ratio, $\sqrt{\left(\frac{8k_B T}{\pi\mu} \right)}$ is the average relative velocity and $e^{\frac{-E_a}{RT}}$ is related to the fraction of collisions with enough energy to react.

TST: This is a probability-based (stat mech) model, that focuses not just on favorability of reactants colliding, but rather, favorability of the transition state relative to the reactants. Note Q = partition function, $L^\ddagger$ is a symmetry factor depending on the molecular geometry of the TS, which is equal to:

$$L^\ddagger = \frac{\frac{1}{\sigma^\ddagger}}{\prod \frac{1}{\sigma_{\text{react}}}}, \text{ where } \sigma \text{ is a symmetry number}$$

$$k_{TST} = L^\ddagger \left(\frac{k_B T}{h} \right) \left(\frac{Q^\ddagger}{\prod(Q_{\text{react}})} \right)$$

$$\text{where } Q = Q_{\text{elec}} * Q_{\text{trans}} * Q_{\text{vib}} * Q_{\text{rot}}$$

$$Q_{\text{trans}} \propto (m, T)^{1.5} = \left(\frac{2\pi m k_B T}{h^2} \right)^{1.5}$$

$$Q_{\text{rot linear}} \propto I, T = \frac{8\pi^2 I k_B T}{h^2}$$

$$Q_{\text{rot nonlinear}} \propto I^{0.5}, T^{1.5} = \pi^{0.5} \frac{(8\pi^2 k_B T)^{1.5}}{h^3} (I_A I_B I_C)^{0.5}$$

$$Q_{\text{vib}} = \prod \frac{1}{1 - e^{-h\nu/k_B T}}$$

Solvent effects:

Hughes-ingold rule: If the TS has a larger charge than reactants, rate increases with solvent polarity and vice versa (governing question: does the solvent aid the TS in an electrostatic sense?). If the net charge is the same, localized charge is better.

Like charges: k_s increases with ϵ because of charge screening effects and vice versa for opposite (free ions will group towards a charge, decreasing its effective charge --> screening)

For like charges, k_s increases with ionic strength (Debye-Huckel)

Enzymes

Michaelis-menten eqn: $V = V_{\text{max}} \frac{[S]}{[S] + K_M}$. Derived using the network: $E + S \xrightleftharpoons[k_{-1}]{k_1} ES \xrightarrow{k_2} E + P$, and corresponding rate calculations.

Enzymes: Enzyme-Substrate binds with product. Competitive inhibition is where inhibitors compete with the substrate. So, there is an $E+I \rightarrow EI$ reaction that competes with the $E+S$ reaction. Anticompetitive binds to the ES complex, so there is a potential of $ES+I \rightarrow EIS$ that competes with $ES \rightarrow P$.

Adsorption/Catalysts

Catalysts, ofc, lower reaction barriers, but what is really happening is that it is taking it through an thermodynamically favorable alternate pathway. For adsorption, Substrate adsorbs on a surface, there's a reaction, then desorption. This is at a much lower energetic level.

1. Adsorption, 2. Reaction of adsorbates, 3. Desorption.

Easier to work with pressures for substrate reactions. θ_A is the coverage, which is # of adsorbed A/# of total sites. θ corresponds to a free site.

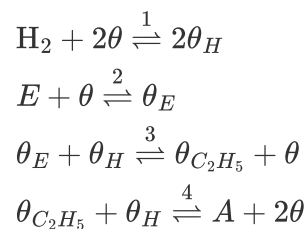
Adsorption can be direct (reaction with surface) or indirect (weak precursor state + chemisorption)

Langmuir isotherm for direct adsorption: $r_{ads} = k_{ads}p_A(1 - \theta_A)$, $r_{des} = k_{des}\theta_A$, so $\theta_A = \frac{K_A p_A}{1 + K_A p_A}$.

In english: $K_A = \frac{k_{ads}}{k_{des}}$ and describes the tendency of a substance to adsorb vs desorb. The rate of adsorption would naturally be corresponding to the free sites, where there is no A, and the amount of A, and desorption would be dependent on how much A is adsorbed.

Take a reaction: $C_2H_4 + H_2 \rightarrow C_2H_6$, let's call this $E + H_2 \rightarrow A$.

Sequence: E adsorbs, H_2 adsorbs and dissociates, E and H (then) EH and H react, then there's desorption:



You can map out the rate equations of this, keeping in mind that $\frac{k_1}{k_{-1}} = K_1$ and so on

When you write out eqns for $\frac{dA}{dt} = r_4 = k_4\theta_{C_2H_5}\theta_H - \frac{k_4}{K_4}A\theta^2$. Last term can be ignored bc we can assume that forward dominates here. But, we need to solve for the intermediate thetas. The general procedure for this is:

1. Assume an RDS (typically the product forming step), and then only consider the forward rxn for that one, it's not in equilibrium.
2. Map out K values. e.g. $K_1 = \frac{\theta_H^2}{(H_2)\theta^2}$. This allows us to then express θ_H as $f(K, \theta)$, $\theta_H = \sqrt{K_1(H_2)}\theta$. Do the same for all three eqns. Sub in knowns as you get them (use lumped terms for simplicity).
3. Do a site balance: $\theta_T = \theta + \sum \theta_i$ for all species. θ_T is the only true constant in this whole problem.
4. With all species thetas in terms of theta, we can solve for theta in terms of theta total.
5. Sub that theta back in and find a rate eqn in terms of partial pressure, θ_T and K's.

Reactor Design:

Key quantities:

$\tau = \frac{V}{v}$, residence time = volume/flow rate. Average time spent for a given particle in a reactor

Conversion, $X = 1 - \frac{F_j}{F_{jo}} = \frac{F_{jo} - F_j}{F_{jo}}$ (less of a component that flows out, more was converted).

Note - base off limiting reactant

Convenient form: $F_j = F_{jo}(1 - X)$, or $F_{jo} = \frac{F_j}{1 - X}$. If volumetric flow rates are constant, then $C_j = C_{jo}(1 - X)$. Rate laws can be manipulated into a form that depends on X if ever needed using this equivalency. Flow rates, like concentrations here, can be related by Stoich.

Always start reactor design problems with a mole balance. Cancel out flow terms for batch, time term for S.S.

Batch reactor - No flow terms - $\frac{dN_j}{dt} = V \frac{dC_j}{dt} = r_j V$, $\frac{dC_j}{dt} = r_j$

For PFR - No mixing - $\frac{dF_j}{dV} = r_j$ or $\frac{dC_j}{d\tau} = r_j$.

For CSTR - perfect mixing: $\frac{dC_A}{dt} = \frac{C_{A0}v_o}{V} - \frac{C_A v}{V} - r$. A few simplifications can be made here, the most important would be re-jigging this in terms of τ .

Volume scaling for idealized reactors - levenspiel plots can be used.

$$CSTR : V = \frac{F_{A0}}{r_A} \big|_{X_f} * (X_f - X_i)$$

$$PFR : V = \int_{X_i}^{X_f} \frac{F_{A0}}{-r_A} dX$$

Concavity of the X vs $\frac{F_{A0}}{-r_A}$ profile determines which reactor is best to use for optimal conversion. For CSTRs in particular, this can result in a lot of volume reduction. Remember that a PFR can be thought of as infinite CSTRs in series. **Think about why it makes perfect sense - Hint: Reimann.**

Additional design consideration: Recycle stream at a ratio of $R = \frac{v_R}{v_o}$. What does this mean?

When we are looking at the PFR itself, we need to consider different flowrates - map conversions and flowrates as a function of recycle stream, do balances around the mixing point.

Real reactors can be conceptualized in terms of idealized reactors:

Modeling Real Reactors

Continuity eqn:

$$\frac{\partial C_j}{\partial t} + \nabla \cdot (C_j \mathbf{u}) + \nabla \cdot \mathbf{J}_j = R_j$$

(In english: transient term + convective term + diffusive flux term = reaction rate term)

In a PFR, you can neglect dispersion terms, assume that the fluid moves as a plug, and for a CSTR, you have perfect mixing, conc is same throughout control volume, and therefore, so is rate

Thermal balance:

$$\sum_j M_j C_j c_{pj} \left(\frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T \right) = \sum_i (-\Delta H_i) r_i + \nabla \cdot (\lambda \nabla T) - \sum_j \mathbf{J}_j \nabla H_j + Q_{\text{rad}}$$

(1)	(2)	(3)	(4)	(5)	(6)
Accumulation	Flow (convection) of sensible heat	Heat of reaction	Conduction	Thermal diffusion	Radiation

B.C. for reactors:

Symmetry/no flux: $\frac{dC}{dr}(r = 0, R) = 0$ (or T for a thermal balance)

Dankwerts: Convection/diffusion or convection/conduction for heat balance:

$$z = 0 : u_z(r) C_{jo} = u_z(r) C_j(z = 0, r) - D(r) \frac{dC_j}{dz} \big|_{z=0}$$

$$\rho_f C_p u_z(r) [T_0 - T(z = 0, r)] = \lambda(r) \frac{dT}{dz} \big|_{z=0}$$

Parallel reactions:

Take reactions: 1: $A \rightarrow B$ and 2: $A \rightarrow C$, where B is desired and C is an undesired byproduct

Map out the rates:

$$r_1 = k_1 C_A^{\alpha_1}$$

$$r_2 = k_2 C_A^{\alpha_2}$$

Selectivity is the ratio of those rates: $r_{\text{main product}}/r_{\text{byproduct}}$, represents the propensity of the main product over the byproduct. Overall and instantaneous selectivity have different definitions:

$$S = \frac{r_1}{r_2} = \frac{k_1}{k_2} C_A^{\alpha_1 - \alpha_2}$$

$$\bar{S} = \frac{F_C}{F_B}$$

For a CSTR, instantaneous and overall selectivity are identical because of perfect mixing

For series/parallel reactions, yield is a measure of how much of the desired product is produced for a given amount of consumed reactant:

$$Y = \frac{r_{\text{desired}}}{-r_{\text{reactant}}}$$

$$\bar{Y} = \frac{F_{\text{desired}}}{F_{\text{reactant},0} - F_{\text{reactant}}}$$

RTD: Residence time distribution, a probability distribution of the time spent by an infinitesimal particle in the control volume - like any PDF:

$$\int_0^\infty E(t) dt = 1$$

And by integrating to or from some value t_1 , you can get the fraction coming before/fraction remaining after t_1 .

Theoretical limits:

CSTR: $E(t) = \frac{1}{\tau} e^{-t/\tau}$.

PFR: $E(t = \tau) = 1$,
 $E(t \neq \tau) = 0$.

How can certain changes impact the overall RTD?

1. Addition of a PFR to a series basically results in an offset by τ . (Multiple PFRs in series: offset adds)
 1. Note: Order of PFR/CSTR doesn't impact RTD, but it can impact conversion (Levenspiel)
2. Multiple CSTRs - shape gets squashed, initial increase in P before decrease
3. PFR with recycle ratio - multiple peaks in $E(t)$, higher recycle ratio leads to a different change in peak intensity
4. Poor mixing in a CSTR - $E(t)$ will skew more - molecules can get mixed away or not mixed enough depending on impeller position. Lower impeller speed will bring us closer to a PFR.
5. Dead volume - Smaller usable volume, so the $\tau_{\text{obs}} < \tau_{\text{actual}}$, so timescale of decay is quicker than τ .
6. Bypass - Smaller observed volumetric flowrate, so the decay timescale will be longer than τ .

7. PFR with eddies - Mixing leads to some elements of a CSTR, small amounts would lead to spreading of the PFR RTD.
8. Channeling - Bimodal distribution, some molecules will have more channel effects than others.

How can we find $E(t)$ without prior knowledge? A: Tracer experiments. For some tracer T with a defined concentration profile:

$$E(t) = \frac{C_T(t)}{\int_0^\infty C_T(t)}$$

Mass transfer in catalyst particle: Transport of reactants from bulk to catalyst surface, transport within the porous catalyst, adsorption of reactants on site, reaction, desorption, transport of products to catalyst surface, and then to bulk.

Depending on the limitation, the rate of the reaction can depend on mass transfer or rate constant. Depending on the limitation, the slope of $\ln(k)$ vs $1/T$ would be different (higher magnitude slope (E_a/R) for rxn limited).

Scaling for porous catalyst slab in first order (steady-state)- derivation of thiele modulus (use this as an example of deriving a generalized Thiele modulus)

$$D_e \frac{d^2 C_A}{dz^2} - k C_A = 0$$

$$B.C.: D_e \frac{dC_A}{dz} \Big|_{center} = 0$$

$$C_A|_L = C_{As} = C_{Ab}, \text{ no external mass transfer limitations}$$

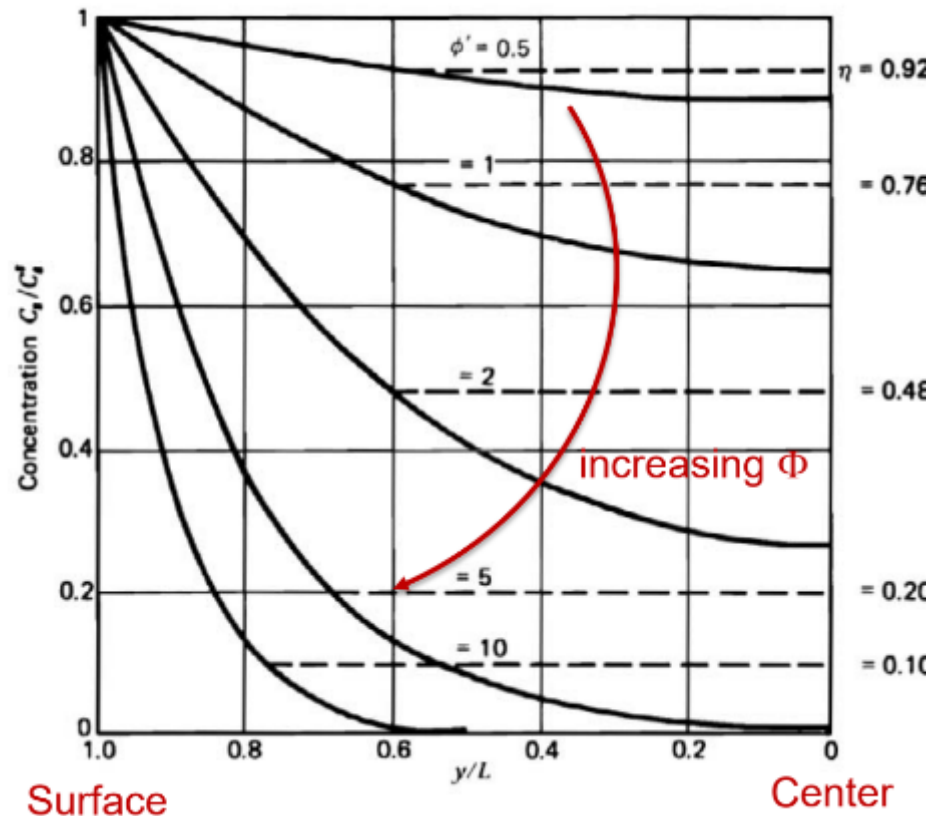
$$\text{Nondimensionalize: } \theta = \frac{C_A}{C_{Ab}}, \xi = \frac{z}{L}$$

$$\frac{d^2 \theta}{d\xi^2} - \left(\frac{kL^2}{D_e} \right) \theta = 0$$

$$\frac{d^2 \theta}{d\xi^2} - \Phi^2 \theta = 0$$

$$\Phi^2 = \left(\frac{kL^2}{D_e} \right), \Phi^2 \text{ is the Thiele modulus}$$

Thiele modulus represents the relative rates of reaction and transport, high thiele modulus means diffusion limited, low means reaction limited.



Thiele modulus value influences concentration profiles - lower Φ^2 means reaction limited, which would result in lower concentration gradient.

Effectiveness factor: η = reaction rate/surface rate. So $r_{obs} = \eta k C_{As}$. With large diffusion limitations, or higher Thiele modulus, you're more likely to have a concentration gradient and lower internal rate the surface rate, so η would be less. And in fact for diffusion-limited situations ($\Phi \gg 1$): $\eta \sim \frac{1}{\Phi}$.

Generalized Thiele:

$$\Phi = \frac{V_p}{S_x} \sqrt{\left(\frac{n+1}{2}\right) \frac{k c_s^{n-1}}{D_e}}$$

V_p/S_x is particle volume/surface area, or characteristic length.

Multiple steady states can come from: Complex kinetics - negative reaction orders, exothermic rxns with external mass transfer limitations. Governing thermal question: does a Q vs T curve have multiple intersections with a heat absorption line? $Q_{gen} = Q_{rem}$. For a given generation curve, the number of intersection points = number of steady states.

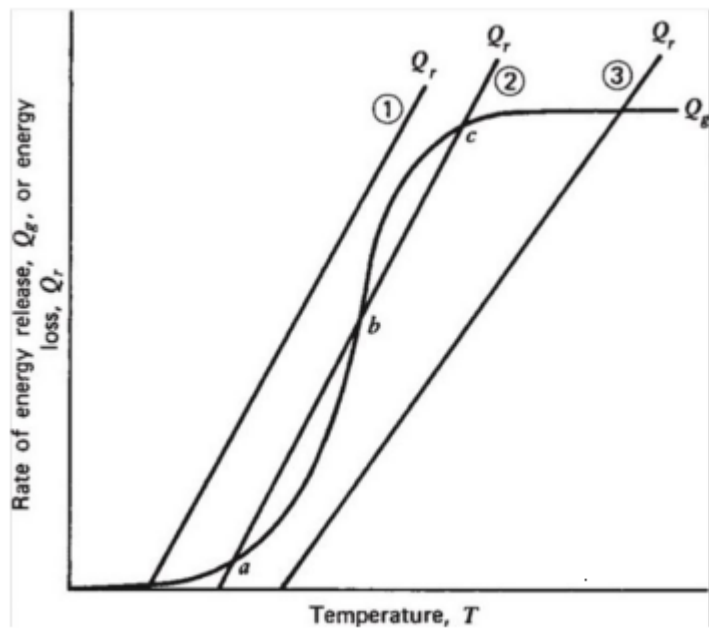


Figure 10.3